

COURSE CODE/TITLE: CPE 010-CPE21S1 - Data Structures and Algorithms	SECTION: CPE21S1
DATE PERFORMED: Sep 15, 2026	DATE SUBMITTED: Sep 15, 2026
NAME: Mauro, Ryan Lewis Medina, Jomz Ryle Mendoza, Roy Gabriel Mendoza, Ryo Shin	INSTRUCTOR: Engr. Jimlord Quejado

ASSIGNMENT NO. 9.1 Implementing Trees

1. PROCEDURE OUTPUT

What procedures did you perform?

1. What is a binary search tree?	A Binary Search Tree (BST) is a type of binary tree used to store and organize data in a specific order. Each node contains a value, with smaller values placed in the left subtree and larger values placed in the right subtree. This arrangement allows operations such as searching, inserting, and deleting data to be performed efficiently, especially when the tree is well-balanced.
2. Where can binary search trees be used?	A Binary Search Tree can be used in applications that need data to be stored in an organized and searchable way. For example, in a Git repository, a BST could be used to organize and search repository-related information such as file names, project records, or other sorted data. If there are many files in a repository, the BST can help locate a specific file by comparing its name and deciding whether to search on the left or right side of the tree. This makes it useful for fast searching and maintaining organized data.
3. What is a tree traversal?	A tree traversal is a systematic process of visiting every node in a tree data structure, like in our binary tree. In the case of our binary tree data structure, we access the root first then its childrens, basically left and right linked list or nodes, which then can contain another set of children with a maximum of 2. We have managed to do this in our code by creating a template based tree data structure that stores the data into each node. We then push the new data or link the new nodes by using our insertleft or insertright functions that technically makes the children of the root, and so on. By following this concept, we could finally know the contents of each nodes, and follow them and outputting the content by using something like this: <code>std::cout << "Left child of A: " << root->getLeftChild()->getRootVal() << '\n';</code>.

4. Differentiate a post-order and pre-order traversal give examples.	In the most simplest terms, both are forms of traversal where pre-order is when you travel starting from the root towards the left subtree, followed by the right. Post-order is just the opposite of that where we travel from the left, then the right subtree, and back to the root
	Example : <pre> graph TD A --> B A --> C B --> D B --> E C --> F </pre> Pre-order: (Root → Left → Right) [A, B, D, E, C, F] Post-order: (Left → Right → Root) [D, E, B, F, C, A]
5. What is a parse tree and its use?	A parse tree is a structure that represents the arrangements and relationships of the parts of an expression or sentence based on specific rules or grammar. In C++/Programming, it is commonly used to represent mathematical and logical expressions. It helps show which operations should be performed first. Parse trees are mainly used in compilers and interpreters to analyze and understand the structure of programming code. They can also help detect errors in an expression and make it easier for a computer to process and evaluate the given input.

Reference:

GeeksforGeeks. (2026, February 16). *Binary search tree*. GeeksforGeeks.

<https://www.geeksforgeeks.org/dsa/binary-search-tree-data-structure/>

2. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: None

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: None

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: None

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.



Please have the honor of pledge in the comment section or at the bottom of your activity document/exam or any form of file submission.



I affirm that I will not give or receive any unauthorized help on this activity/exam and that all work will be my own.